

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

STG213 - R

Amazon S3 encryption best practices

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AWS



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Agenda

- Flexible options to encrypt your data
- Applying and enforcing encryption
- Auditing and remediating non-compliant encryption



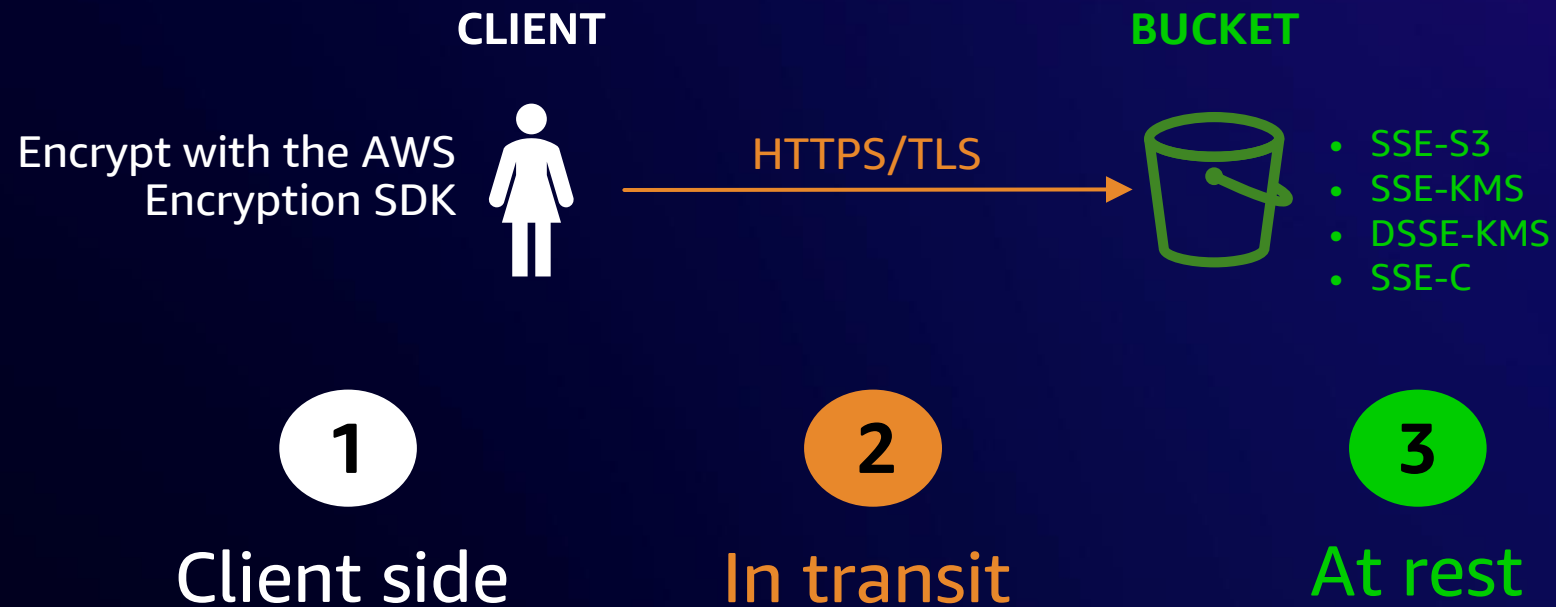
Amazon S3 is secure by default

FROM BEST PRACTICES TO DEFAULTS: AMAZON S3 SIMPLIFIES SECURITY



Flexible options to encrypt your data

Amazon S3 encryption support

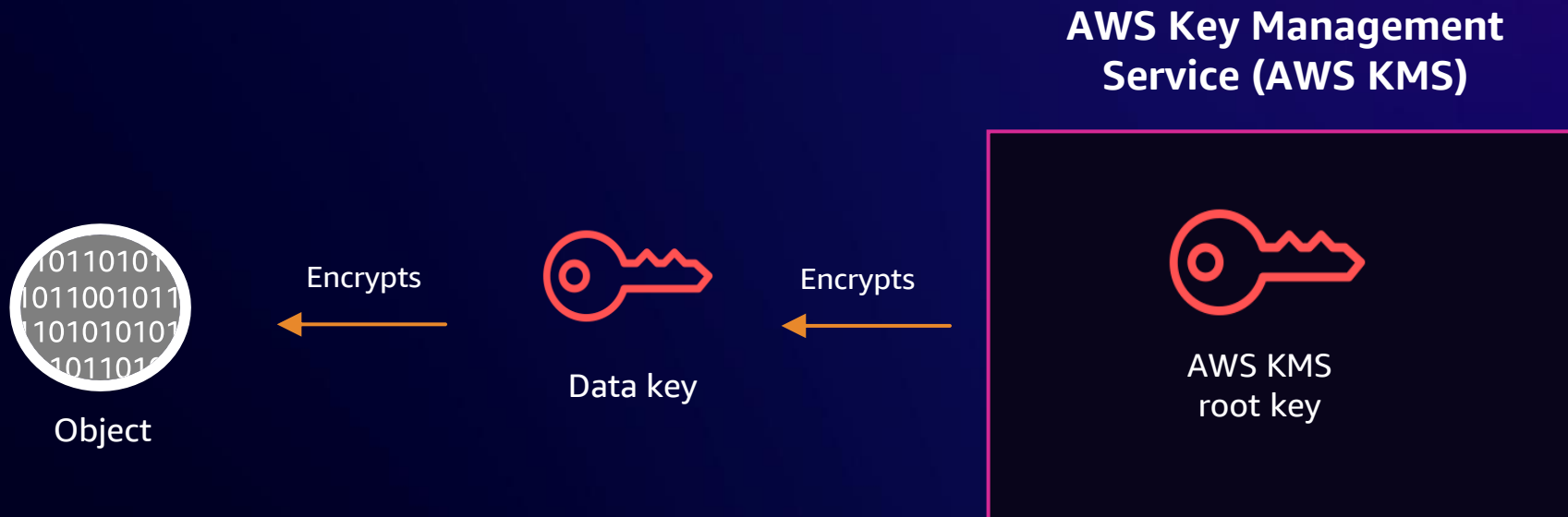


Amazon S3 encryption at rest options

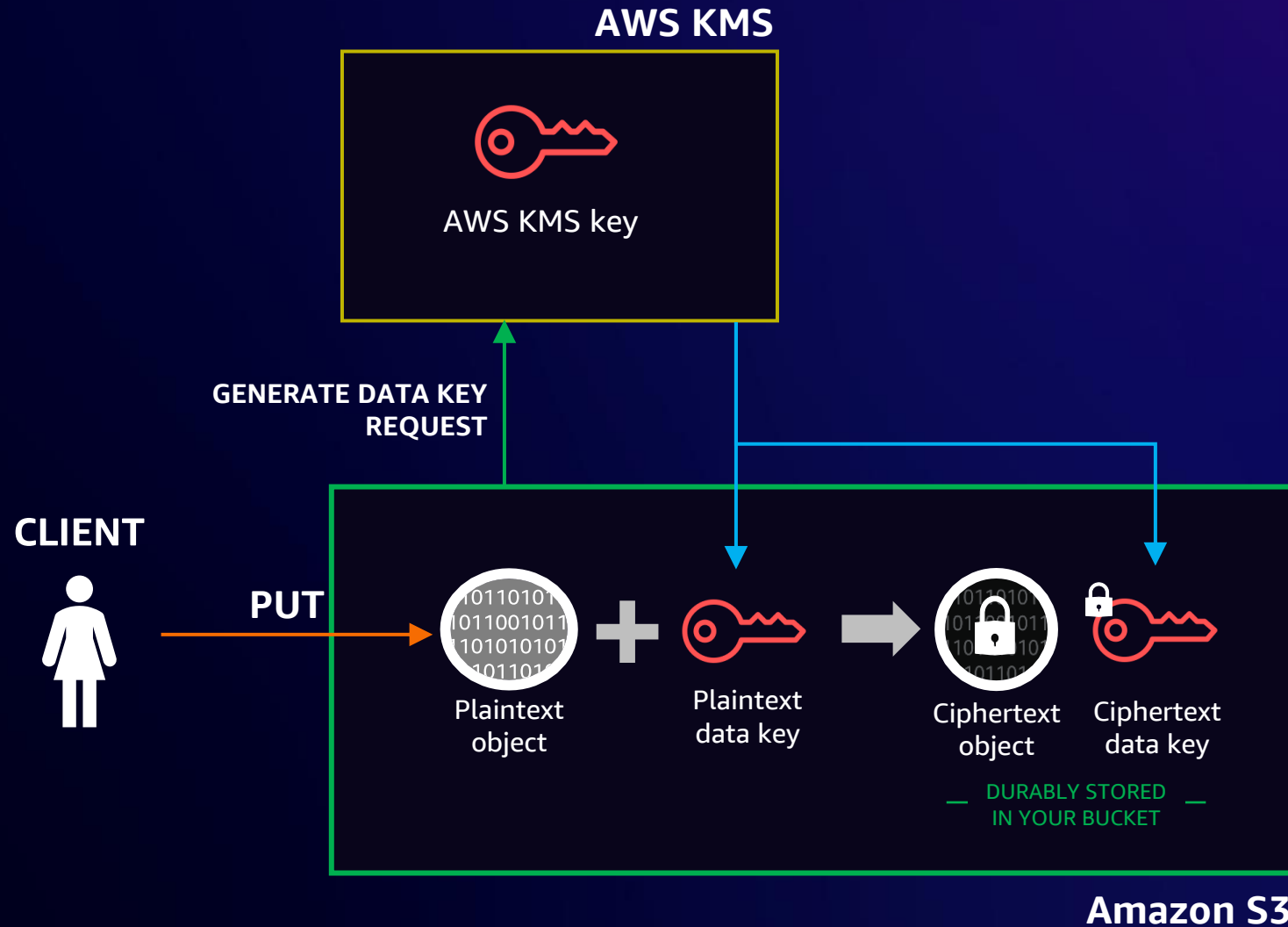
NEW

	Amazon S3 managed keys (SSE-S3)	AWS KMS keys (SSE-KMS) with AWS managed key	AWS KMS keys (SSE-KMS) with customer managed key	Dual-layer SSE with AWS KMS keys (DSSE-KMS)
Default behavior	NEW! As of Jan. 2023 for all new objects	✗	✗	✗
Data shareability	✓	✗	✓	✓
Key rotation	Rotated regularly by Amazon S3	Rotated regularly by AWS	✓	✓
AWS CloudTrail logs for key usage	✗	✓	✓	✓
Manage key policies	✗	✗	✓	✓

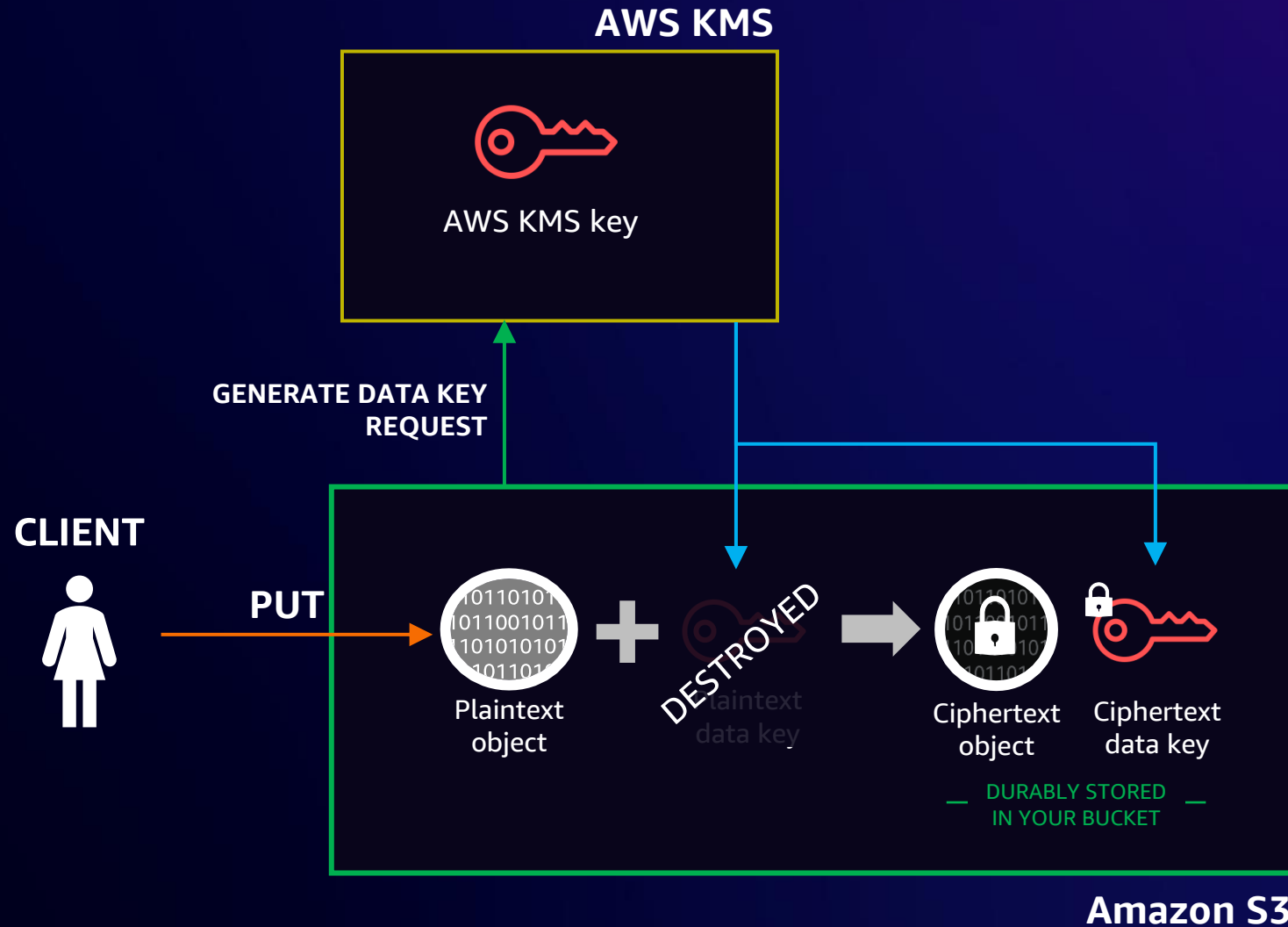
Envelope encryption



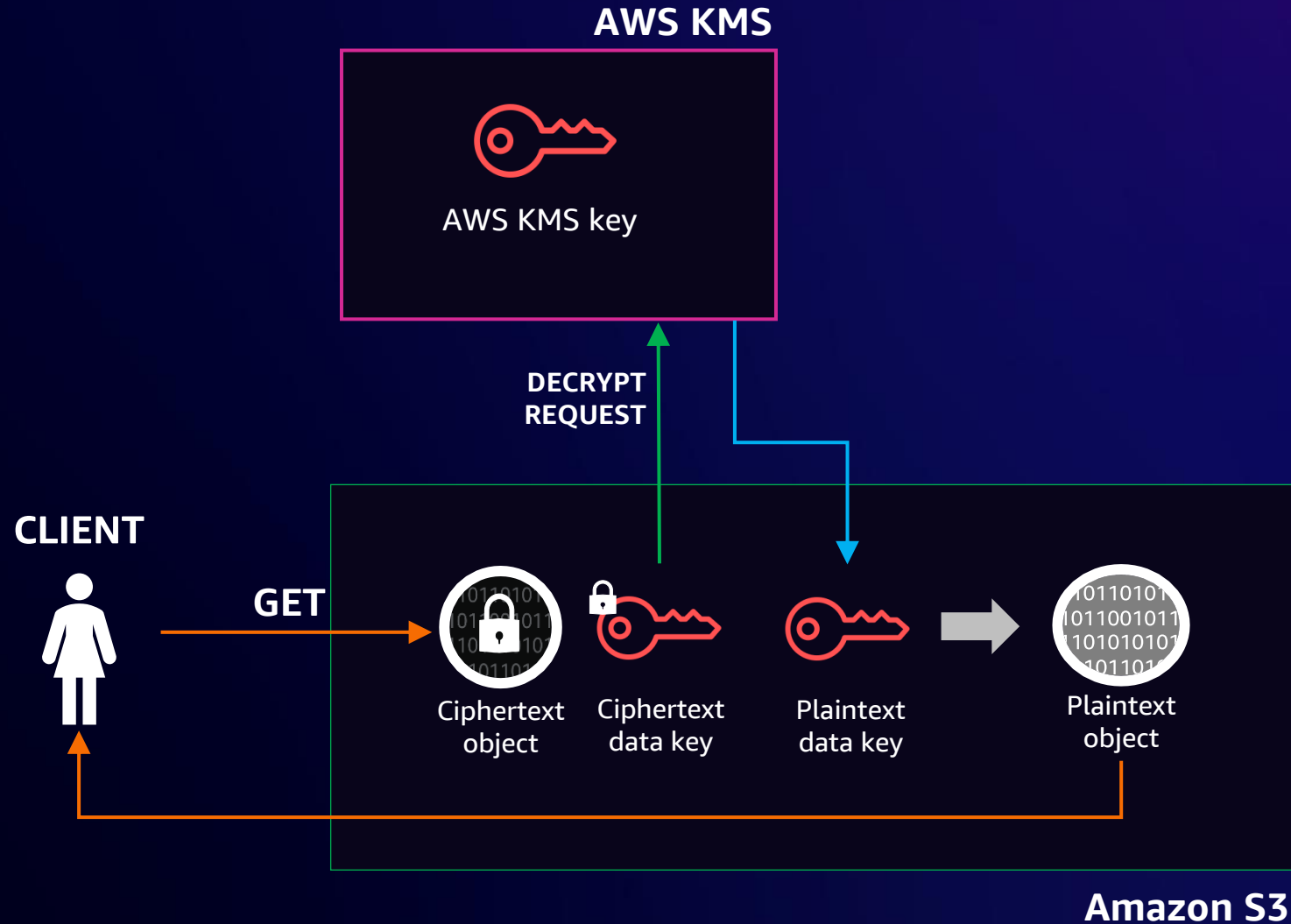
How an encrypt request works with AWS KMS



How an encrypt request works with AWS KMS



How a decrypt request works with AWS KMS



Amazon S3 Bucket Keys



Increase
performance
for encryption



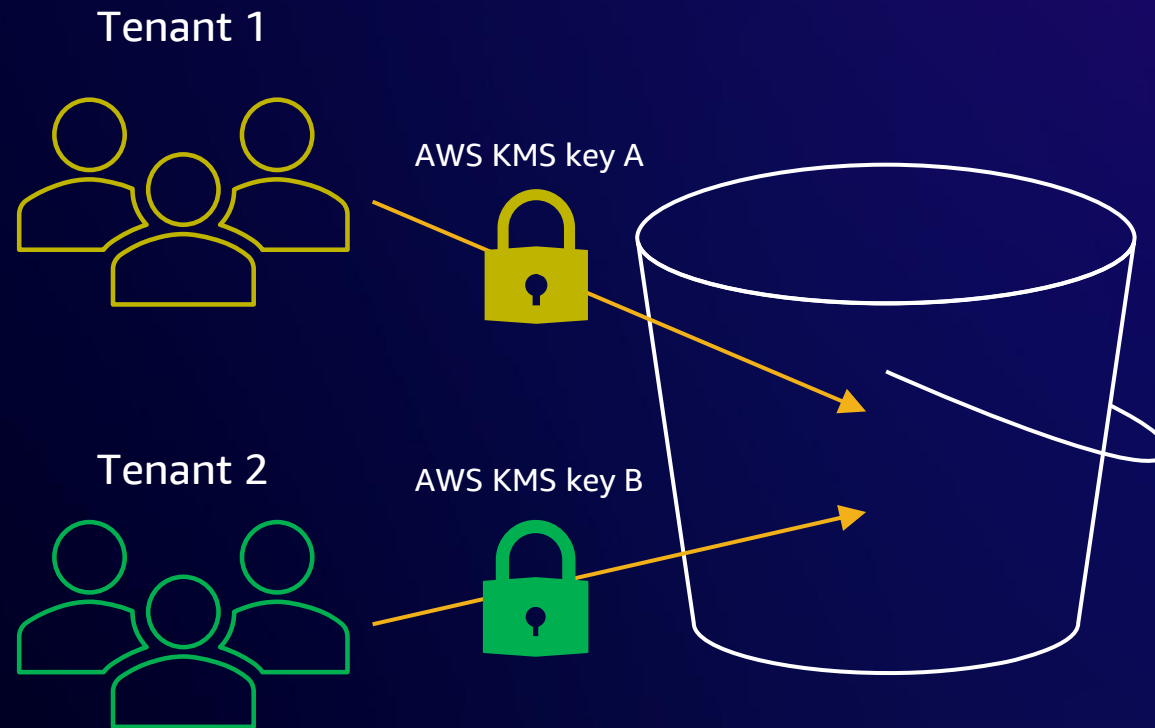
Reduce request costs for
AWS KMS-backed
server-side encryption by

UP TO 99%

Whiteboard: How an encrypt request works with Amazon S3 Bucket Keys

- What is a bucket-level key?
- How does it create unique data keys?
- Do Amazon S3 Bucket Keys work on GetObject requests?

Amazon S3 Bucket Keys in multi-tenant environments



Whiteboard: How encrypt requests work with S3 Bucket Keys in a multi-tenant bucket

- What happens if I have multiple different customer managed keys?
- Can I have multiple different bucket-level keys in my bucket simultaneously?

Applying and enforcing encryption in Amazon S3

Specifying encryption at the object level



SSE-S3

```
"x-amz-server-side-encryption": "AES256"
```

SSE-KMS

```
"x-amz-server-side-encryption": "aws:kms"
```

```
"x-amz-server-side-encryption-aws-kms-key-id": "your KMS key ID"
```

```
"x-amz-server-side-encryption-bucket-key-enabled": "true/false"
```

DSSE-KMS

```
"x-amz-server-side-encryption": "aws:kms:dsse"
```

```
"x-amz-server-side-encryption-aws-kms-key-id": "your KMS key ID"
```

NEW

Amazon S3 default encryption configuration

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

- ☐ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☒ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on DSSE-KMS pricing, see DSSE-KMS encryption pricing on the [Amazon S3 pricing page](#). [↗](#)

AWS KMS key [Info](#)

- ☐ Choose from your AWS KMS keys
- ☒ Enter AWS KMS key ARN

AWS KMS key ARN

[Create a KMS key](#) [↗](#)

Format (using key id): arn:aws:kms:<region>:<account-ID>:key/<key-id>
(using alias): arn:aws:kms:<region>:<account-ID>:alias/<alias-name>

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#) [↗](#)

- ☐ Disable
- ☒ Enabled

One-time bucket-level setup

Use SSE-S3 by default or select a key management strategy

Specify your AWS KMS customer managed key

Enable Amazon S3 bucket keys!

Enforcing encryption in Amazon S3

USING AWS IDENTITY AND ACCESS MANAGEMENT (IAM) POLICY SYNTAX

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::/sample-bucket-reinvent/*",
      "Principal": {"AWS": "111111111111"}
    }
  ]
}
```

IAM policy

Bucket policy (requires principal statement)

Service Control Policy (SCP)

Use Amazon S3 bucket policy to enforce DSSE

Unless the object will be encrypted at rest with DSSE-KMS, reject the attempt to write it



```
{
  "Effect": "Deny",
  "Action": "s3:PutObject",
  "Resource": "arn:aws:s3:::DOC-EXAMPLE-BUCKET/*",
  "Principal": "*",
  "Condition": {
    "StringNotEquals": {
      "s3:x-amz-server-side-encryption": "aws:kms:dsse"
    }
  }
}
```

Use Amazon S3 bucket policy to enforce a specific AWS KMS key

Unless the object will be encrypted at rest with specified customer managed AWS KMS key, reject the attempt to write it



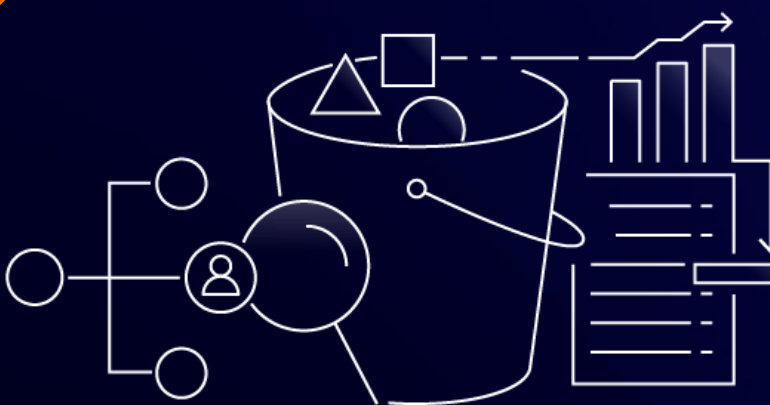
```
{
  "Effect": "Deny",
  "Action": "s3:PutObject",
  "Resource": "arn:aws:s3:::DOC-EXAMPLE-BUCKET/*",
  "Principal": {"AWS": "111111111111"},
  "Condition": {
    "ArnNotEqualsIfExists": {
      "s3:x-amz-server-side-encryption-aws-kms-key-id":
        "arn:aws:kms:us-east-2:111122223333:key/01234567-89ab-cdef-
        0123-456789abcdef"
    }
  }
}
```

Auditing encryption status

Insights and analytics on Amazon S3

Recommended

S3 Storage Lens



Organization-wide visibility into pre-aggregated usage and activity metrics, including encryption details

Free and paid versions available

Amazon S3 Inventory



Object-level metadata like encryption mode

AWS CloudTrail logs



Amazon S3 requests and AWS KMS request logging

Amazon S3 Storage Lens

A storage analytics service that provides organization-wide visibility into storage usage and activity to improve cost efficiency and data protection



1. Drill down by AWS Region, storage class, bucket, and prefix
2. Granular usage and activity metrics
3. Go see our S3 Storage Lens chalk talk (STG329)

Free (5)

Encryption

- Encrypted bytes
- Encrypted object count
- Unencrypted bytes
- Unencrypted object count
- AWS KMS enabled buckets

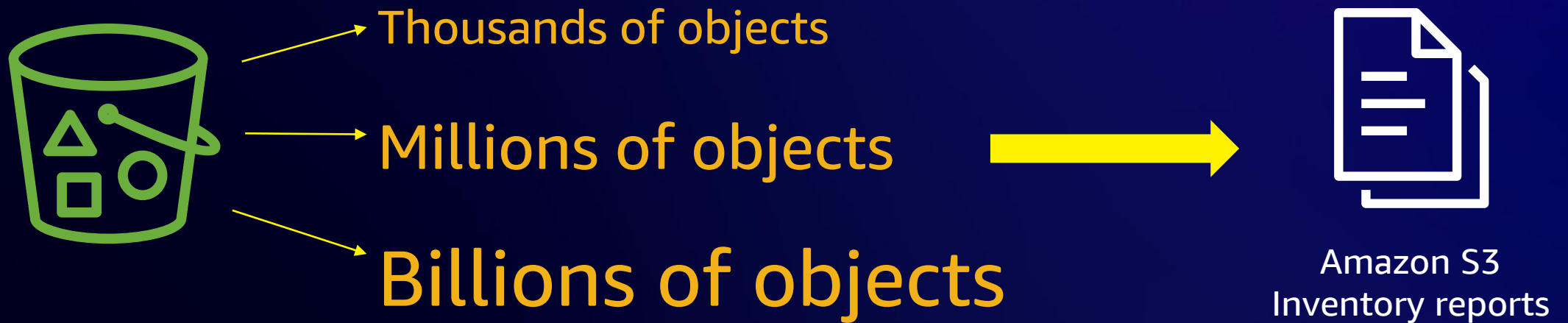
Advanced (1)

Encryption

- AWS KMS requests

Amazon S3 Inventory for object-level insights

Do you need to gain encryption insights about the objects in your buckets?



AWS CloudTrail for key usage insights

- Detailed records for the requests made to your bucket



- 1) Server-side encryption headers for PUT/COPY
- 2) AWS KMS API calls

- Useful in security and access audits
- Ability to query and analyze requests

Encryption auditing tools

	Granularity level	Updating frequency	Price
S3 Storage Lens	Organization Account Bucket Prefix	Daily	\$0.20/mil objects/month
S3 Inventory	Object	Daily/weekly	\$0.0025/mil objects
AWS CloudTrail logs	Request	~5 minutes	\$0.10 per 100,000 data events delivered
Amazon Macie	Bucket Object	~15 minutes	\$0.10 per bucket \$1.00 per GB \$0.010 per 100K objects
AWS Config	Bucket	Adjustable (1 hour–24 hours)	\$0.001 per rule evaluation per Region

Whiteboard: How to identify and re-encrypt non-compliant data

- What's the best way to identify unencrypted data?
- How can I re-encrypt data?
- Should I re-encrypt from SSE-KMS to use S3 Bucket Keys?

Key takeaways

Encryption by default for all new objects

Flexible options to encrypt your data

Enable **Amazon S3 Bucket Keys** to save on AWS KMS cost

Use **bucket policies** to enforce the use of encryption

Audit and remediate your encryption modes with AWS tooling

Continue your AWS storage learning

**Build a
learning plan**



Set your AWS Storage Learning Plans via
AWS Skill Builder

**Increase your
knowledge**



Use **AWS Ramp-Up Guides**
to build your storage
knowledge

**Earn AWS
storage badges**



Demonstrate your
knowledge by achieving
digital badges

aws.training/storage

Thank you!

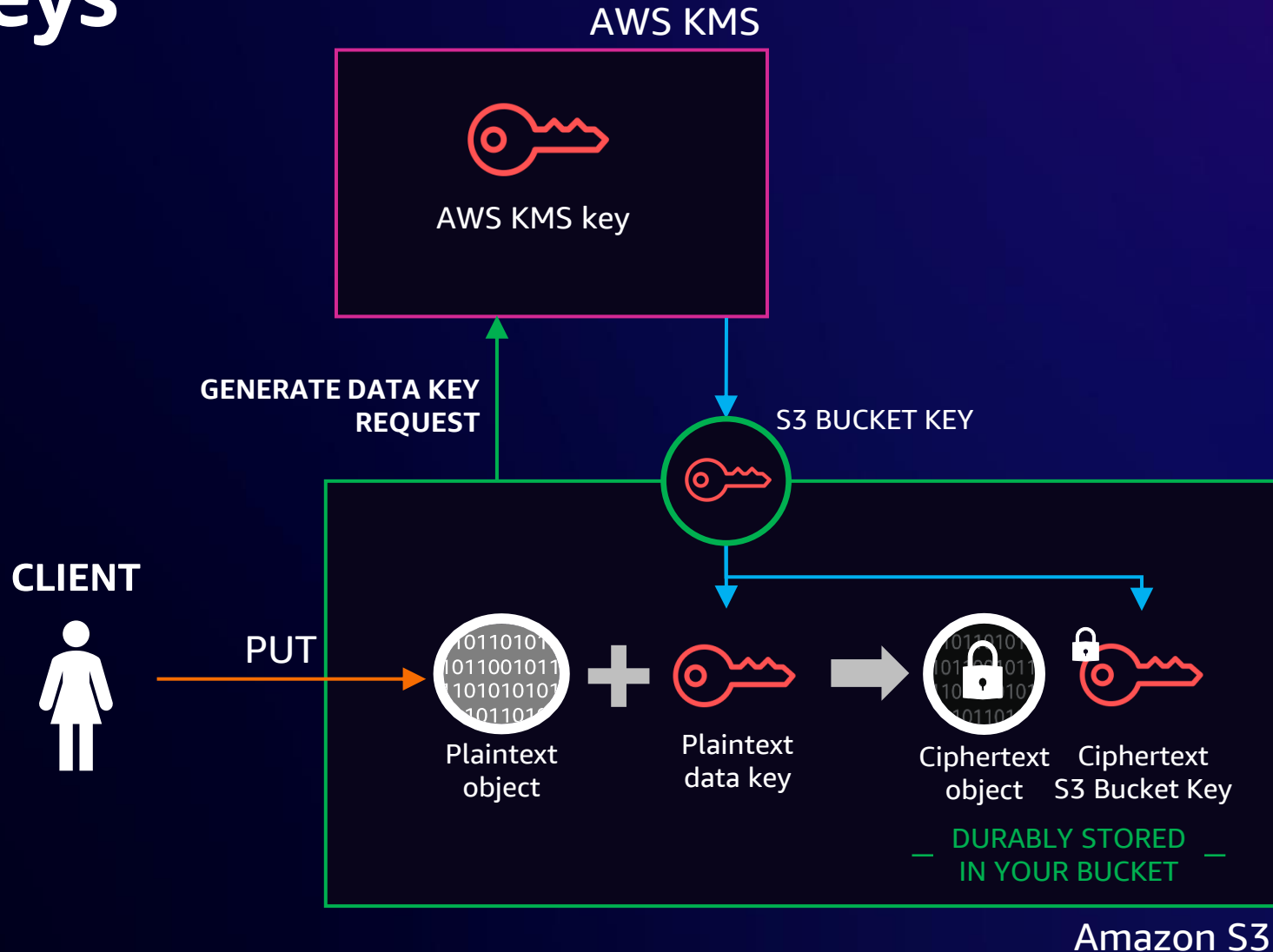


Please complete the session
survey in the mobile app

Backup slides for chalk talk questions

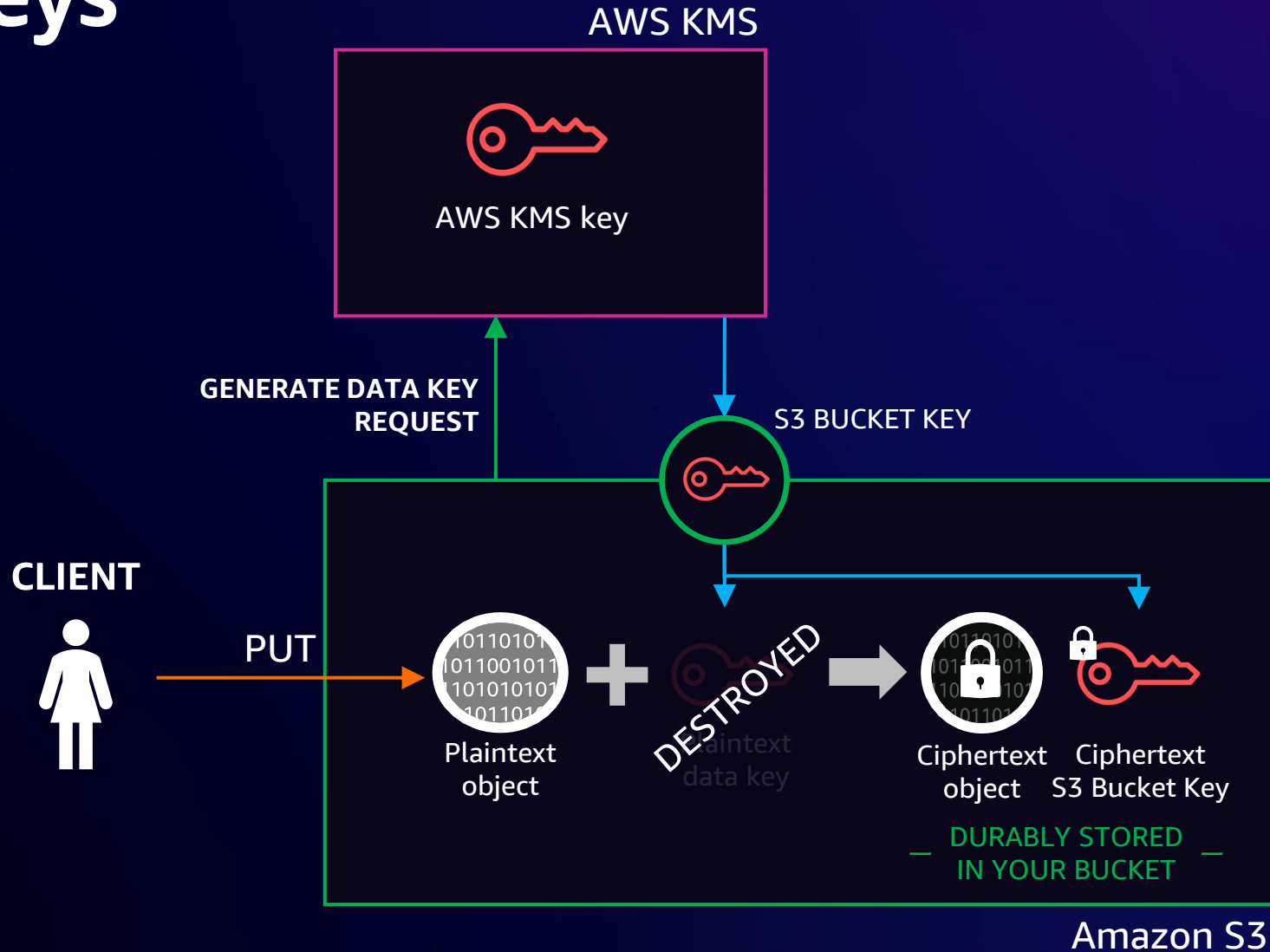
How an encrypt request works with Amazon S3 Bucket Keys

FIRST REQUEST



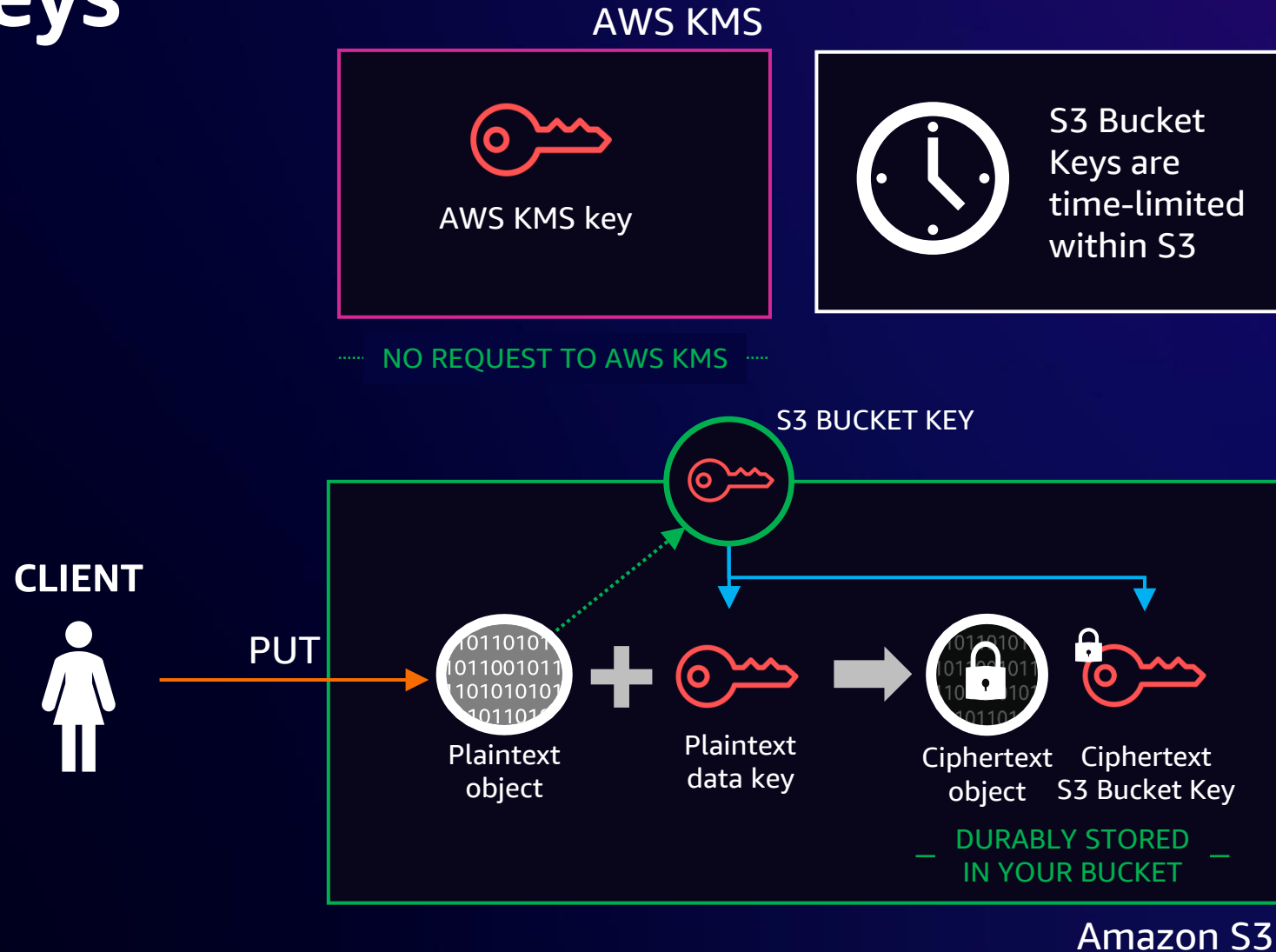
How an encrypt request works with Amazon S3 Bucket Keys

FIRST REQUEST



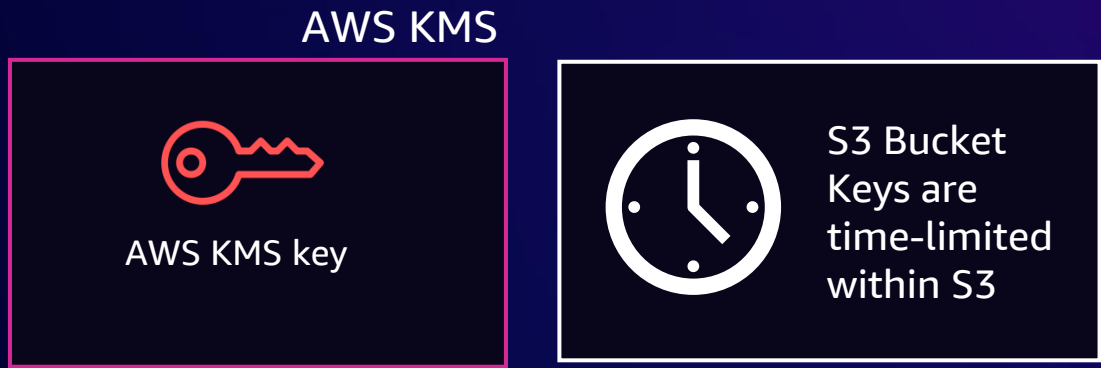
How an encrypt request works with Amazon S3 Bucket Keys

NTH REQUEST

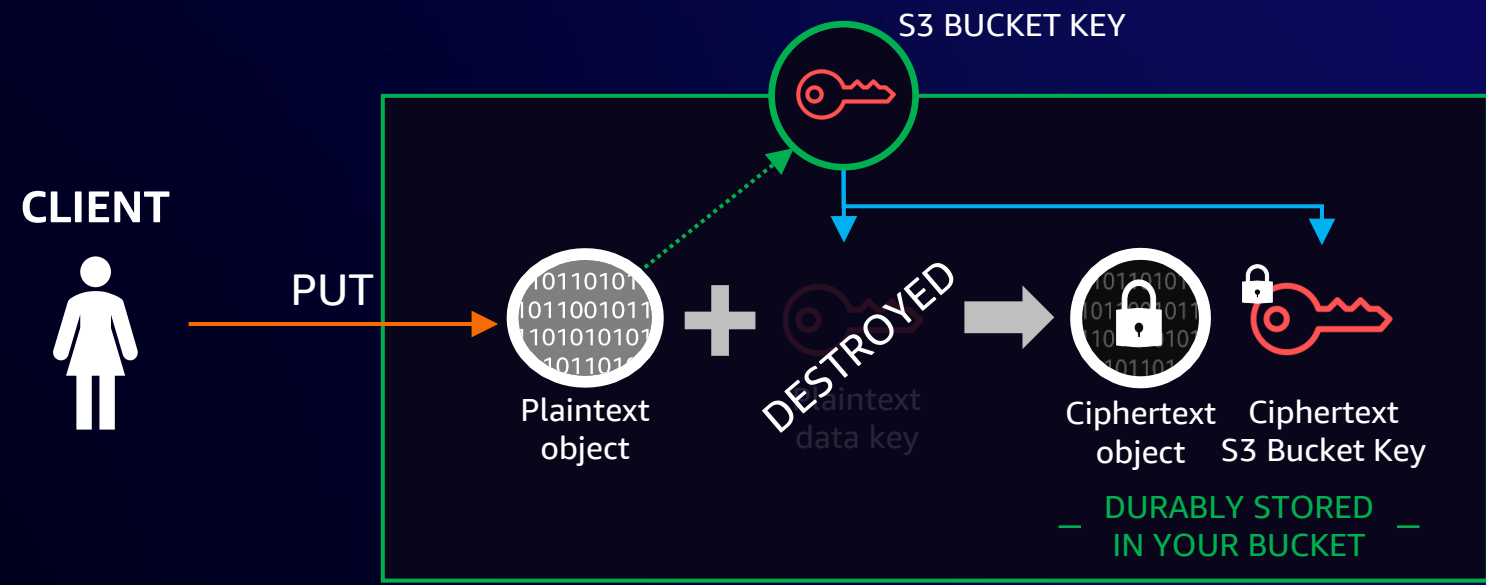


How an encrypt request works with Amazon S3 Bucket Keys

NTH REQUEST



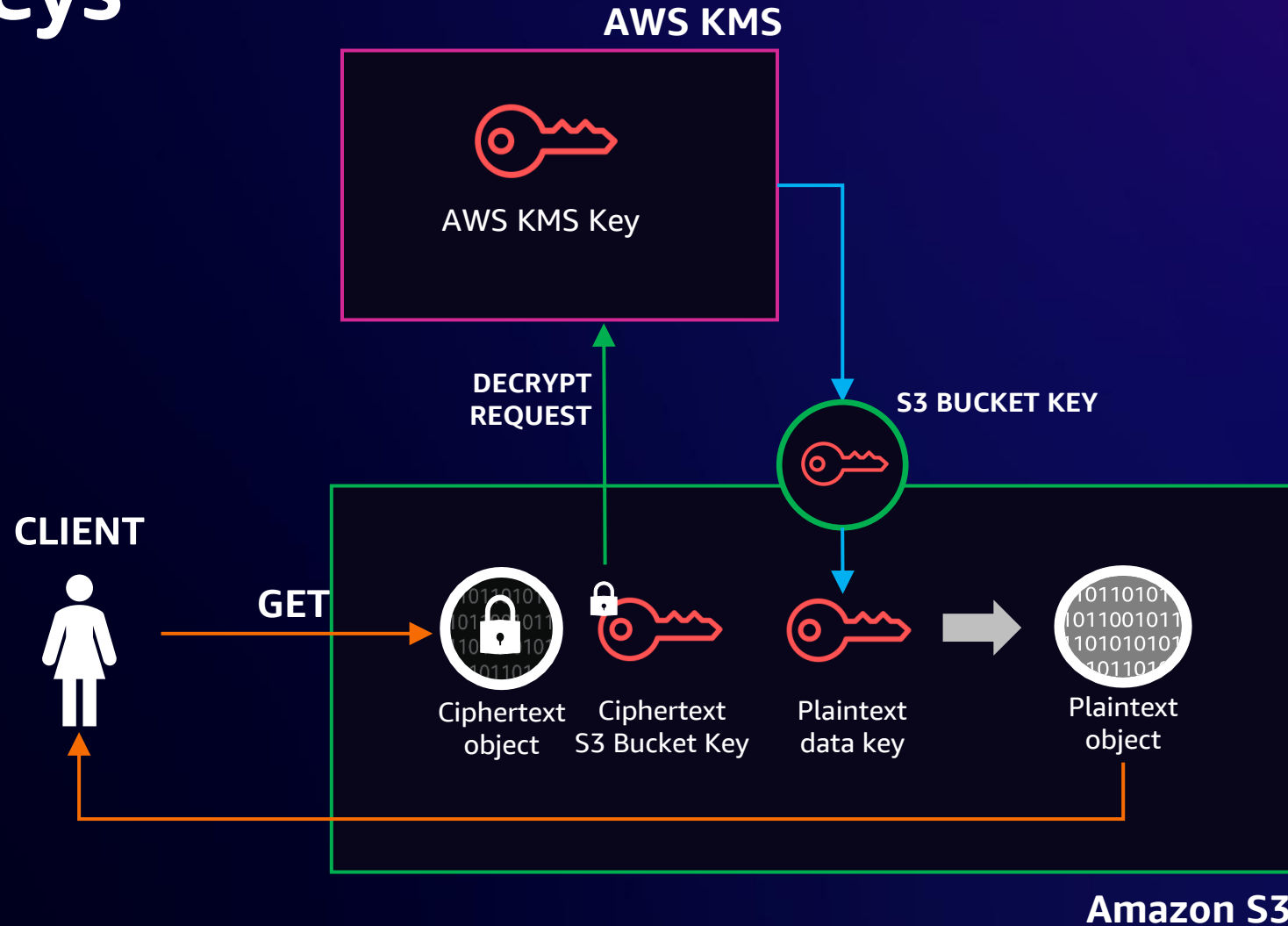
..... NO REQUEST TO AWS KMS



Amazon S3

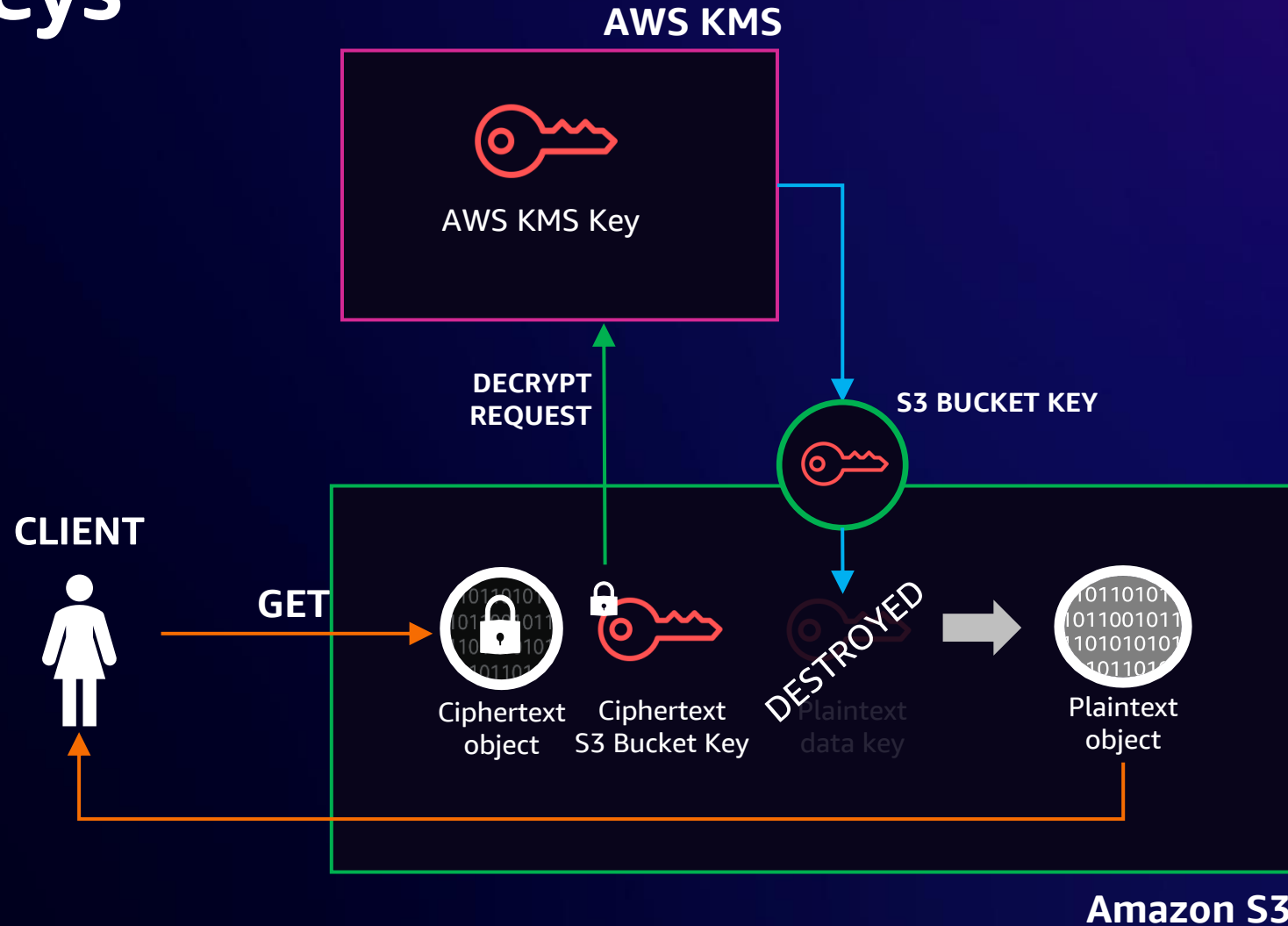
How a decrypt request works with Amazon S3 Bucket Keys

FIRST REQUEST



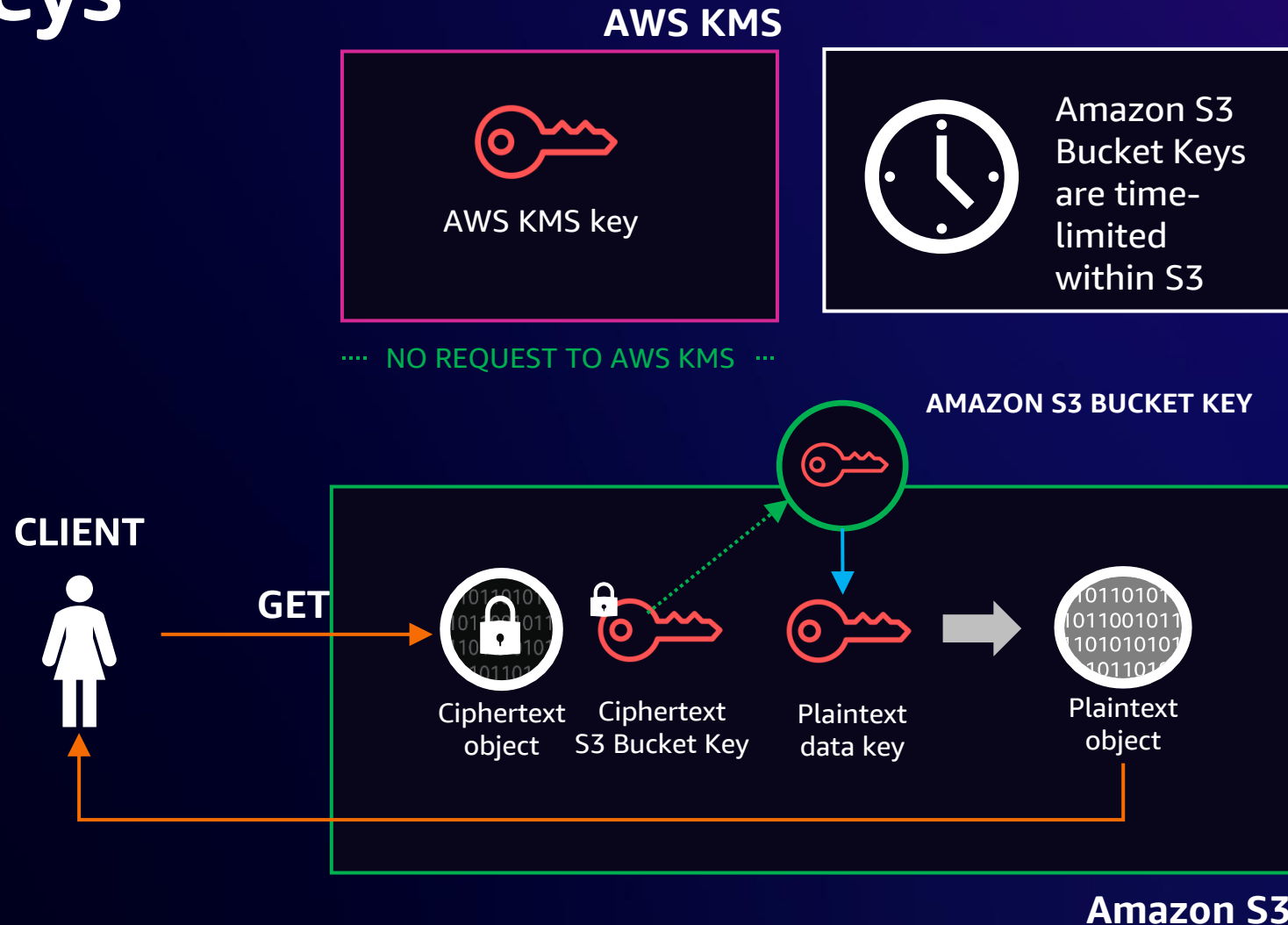
How a decrypt request works with Amazon S3 Bucket Keys

FIRST REQUEST



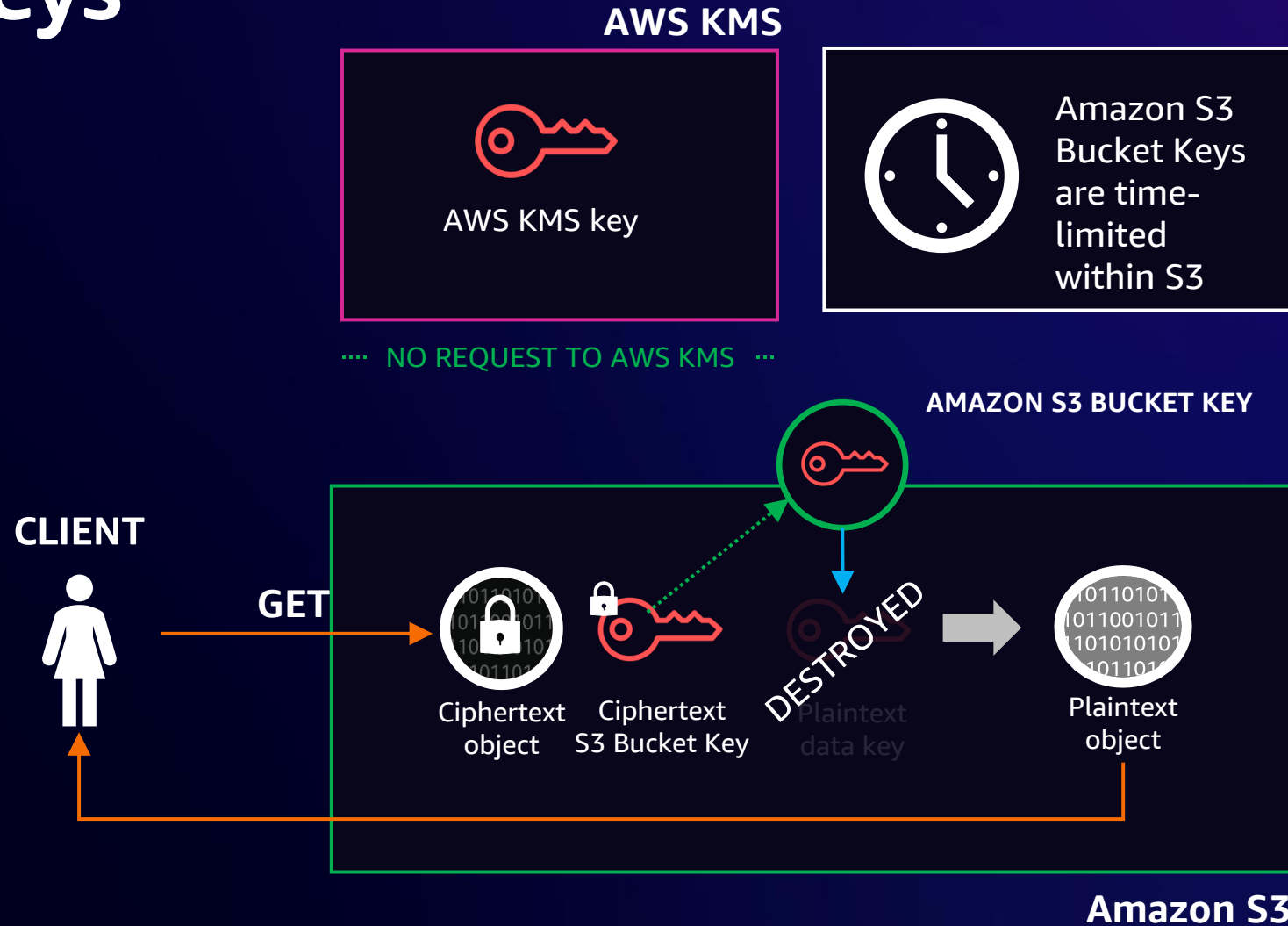
How a decrypt request works with Amazon S3 Bucket Keys

NTH REQUEST



How a decrypt request works with Amazon S3 Bucket Keys

NTH REQUEST



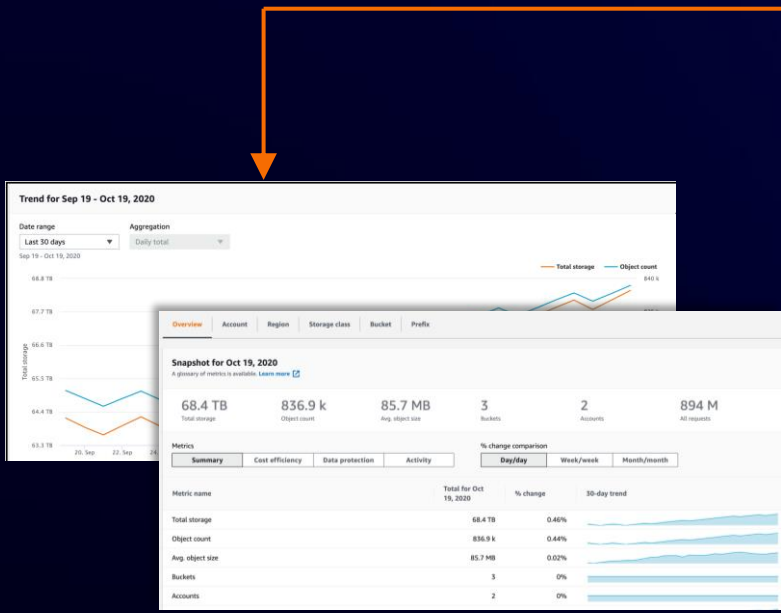
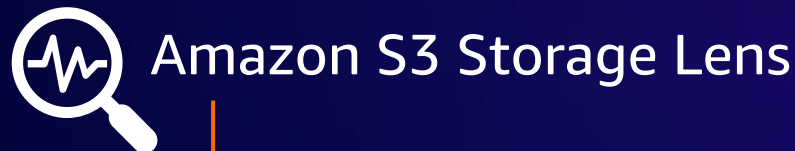
Use Amazon S3 bucket policy to enforce SSE-C

Unless the object will be encrypted at rest with SSE-C, reject the attempt to write it



```
{  
  "Effect": "Deny",  
  "Action": "s3:PutObject",  
  "Resource": "arn:aws:s3:::DOC-EXAMPLE-BUCKET/*",  
  "Condition": {  
    "Null": {  
      "s3:x-amz-server-side-encryption-customer-algorithm":  
        "true"  
    }  
  }  
}
```

Multiple ways to access Amazon S3 Storage Lens data

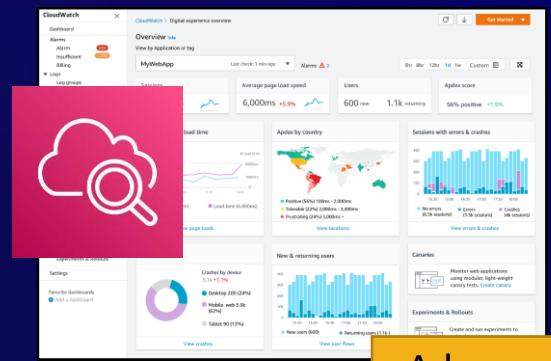


Integrated dashboard
(Amazon S3 console)



ws_account_number	aws_region	storage_class	record_type	record_value	bucket_name	metric_name	metric_value
3.58E+11	us-east-1	STANDARD	BUCKET	StorageBytes	test-bucket	StorageBytes	60214
3.58E+11	us-east-1	STANDARD	BUCKET	ObjectCount	test-bucket	ObjectCount	15
3.58E+11	us-east-1	STANDARD	BUCKET	ReplicatedObjectCount	test-bucket	ReplicatedObjectCount	0
3.58E+11	us-east-1	STANDARD	BUCKET	EncryptedObjectCount	test-bucket	EncryptedObjectCount	0
3.58E+11	us-east-1	STANDARD	BUCKET	EncryptedObjectCount	test-bucket	EncryptedObjectCount	0
3.58E+11	us-east-1	STANDARD	BUCKET	DeleteMarkerObjectCount	test-bucket	DeleteMarkerObjectCount	0
3.58E+11	us-east-1	STANDARD	BUCKET	ObjectLockEnabledStorageBytes	test-bucket	ObjectLockEnabledStorageBytes	0
3.58E+11	us-east-1	STANDARD	BUCKET	ObjectLockEnabledObjectCount	test-bucket	ObjectLockEnabledObjectCount	0
3.58E+11	us-east-1	STANDARD	BUCKET	CurrentVersionObjectCount	test-bucket	CurrentVersionObjectCount	60214
3.58E+11	us-east-1	STANDARD	BUCKET	NonCurrentVersionStorageBytes	test-bucket	NonCurrentVersionStorageBytes	0
3.58E+11	us-east-1	STANDARD	BUCKET	NonCurrentVersionObjectCount	test-bucket	NonCurrentVersionObjectCount	0
3.58E+11	us-east-1	STANDARD	BUCKET	StorageBytes	test-bucket	StorageBytes	495544796
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	ObjectCount	test-bucket	ObjectCount	3004
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	ReplicatedObjectCount	test-bucket	ReplicatedObjectCount	0
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	EncryptedObjectCount	test-bucket	EncryptedObjectCount	0
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	EncryptedObjectCount	test-bucket	EncryptedObjectCount	495544796
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	ObjectLockEnabledStorageBytes	test-bucket	ObjectLockEnabledStorageBytes	3004
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	ObjectLockEnabledObjectCount	test-bucket	ObjectLockEnabledObjectCount	0
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	ObjectLockEnabledObjectCount	test-bucket	ObjectLockEnabledObjectCount	0
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	CurrentVersionObjectCount	test-bucket	CurrentVersionObjectCount	495544796
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	NonCurrentVersionStorageBytes	test-bucket	NonCurrentVersionStorageBytes	3004
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	NonCurrentVersionObjectCount	test-bucket	NonCurrentVersionObjectCount	0
3.58E+11	us-east-1	DEEP_ARCHIVE	ACCOUNT	NonCurrentVersionObjectCount	test-bucket	NonCurrentVersionObjectCount	0

Metrics export
(daily to Amazon S3 bucket)



Advanced
Amazon CloudWatch
(dashboards, API)

“Which buckets have the highest number of unencrypted objects?”

Buckets (10)

You can view an export of data in this dashboard at your [metrics export destination](#). A glossary of metrics is available. [Learn more](#)

Metrics categories

Choose metrics categories

Data protection X

Search buckets

Bucket name	Encrypted object count	% encrypted objects	Unencrypted object count	% unencrypted objects
cloudt reinforce-sample-bucket pijw5poz-benrgau-uu-not-ueete	58.4 k	36.74%	100.5 k	63.26%
reinforce-sample-bucket	58.4 k	36.74%	100.5 k	63.26%
kahlotestbucket2testwill	478	100.00%	-	-
willbucketkahlotest	477	100.00%	-	-
willcavinbucket2	444	99.55%	2	0.45%
willcavinbucket31	18	94.74%	1	5.26%
willcavin-s3secondchance	4	100.00%	-	-
willtest1234jhshsdhd	1	100.00%	-	-
do-not-delete-gatedgarden-audit- [redacted]	-	-	-	-
willcavinbucket54	-	-	-	-

“How can I identify discrepancies in encryption configuration?”

Buckets (30)

You can view an export of data in this dashboard [by configuring a metrics export](#). A glossary of metrics is available. [Learn more](#)

Metrics categories

Choose metrics categories

Data protection X

Search buckets



< 1 2 3 > ⚙

Bucket name	Encrypted bytes	Encrypted object count	Unencrypted bytes	Unencrypted object count	SSE-KMS enabled bucket count	All SSE-KMS requests
demo-2022-5	16.0 GB	9	-	-	-	-
demo-2022-6	45.7 GB	119	-	-	-	-
demo-2022-7	183.2 GB	102	-	-	1	723.3 M
gameday-test-delegateadmin	2.5 GB	4.4 k	-	-	-	2
gameday-test-orgowner	935.4 MB	4.4 k	91.9 MB	1.4 k	-	2
lenz-prod-verification-af-south-1	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-northeast-1	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-northeast-2	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-south-1	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-southeast-1	-	-	9.0 MB	34	-	-

“Should I re-encrypt my objects with Amazon S3 Bucket Keys?”

Buckets (30)

You can view an export of data in this dashboard [by configuring a metrics export](#). A glossary of metrics is available. [Learn more](#)

Metrics categories

Choose metrics categories

Data protection X

Search buckets



< 1 2 3 > ⚙

Bucket name	Encrypted bytes	Encrypted object count	Unencrypted bytes	Unencrypted object count	SSE-KMS enabled bucket count	All SSE-KMS requests
demo-2022-5	16.0 GB	9	-	-	-	-
demo-2022-6	45.7 GB	119	-	-	-	-
demo-2022-7	183.2 GB	102	-	-	1	723.3 M
gameday-test-delegateadmin	2.5 GB	4.4 k	-	-	-	2
gameday-test-orgowner	935.4 MB	4.4 k	91.9 MB	1.4 k	-	2
lenz-prod-verification-af-south-1	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-northeast-1	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-northeast-2	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-south-1	-	-	9.0 MB	34	-	-
lenz-prod-verification-ap-southeast-1	-	-	9.0 MB	34	-	-

AWS Config - Overview



AWS Config

Record and normalize the changes into a consistent format



Manage

Discover resources, record configurations, understand relationships, and capture changes



Evaluate

Check resource compliance with custom and managed AWS Config rules before and after provisioning



Simplify

Use conformance packs to more easily deploy multiple rules and remediations across an account or AWS Region

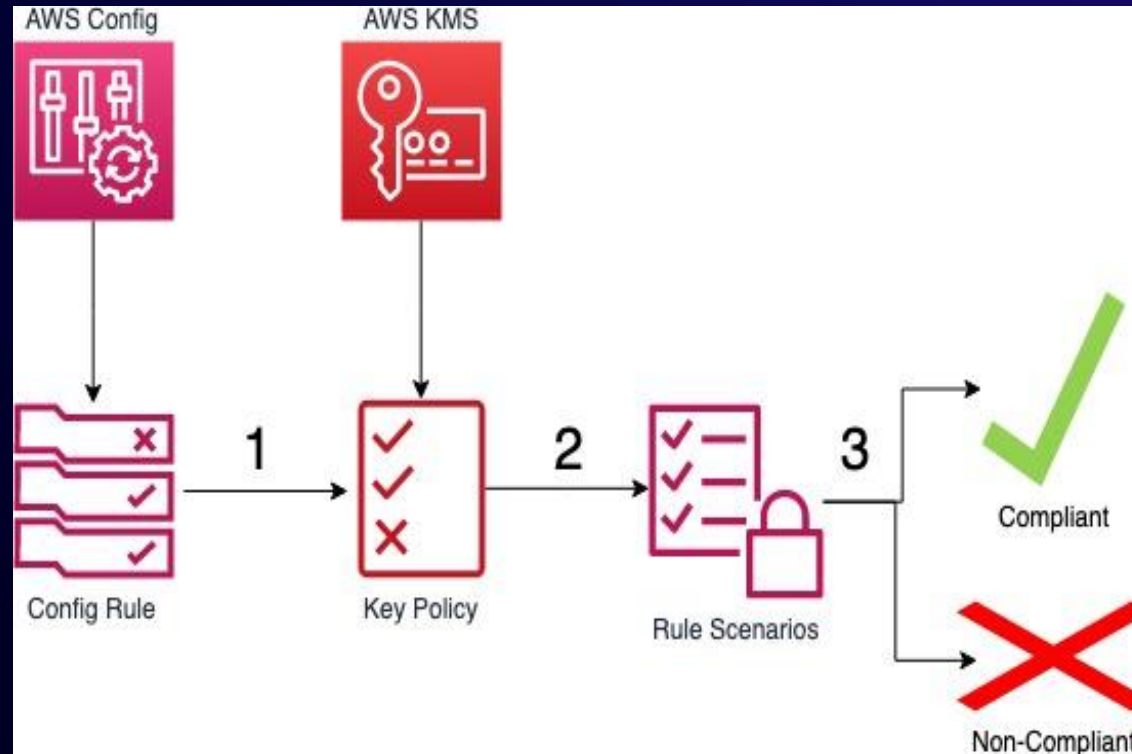
Operational troubleshooting

Compliance and auditing

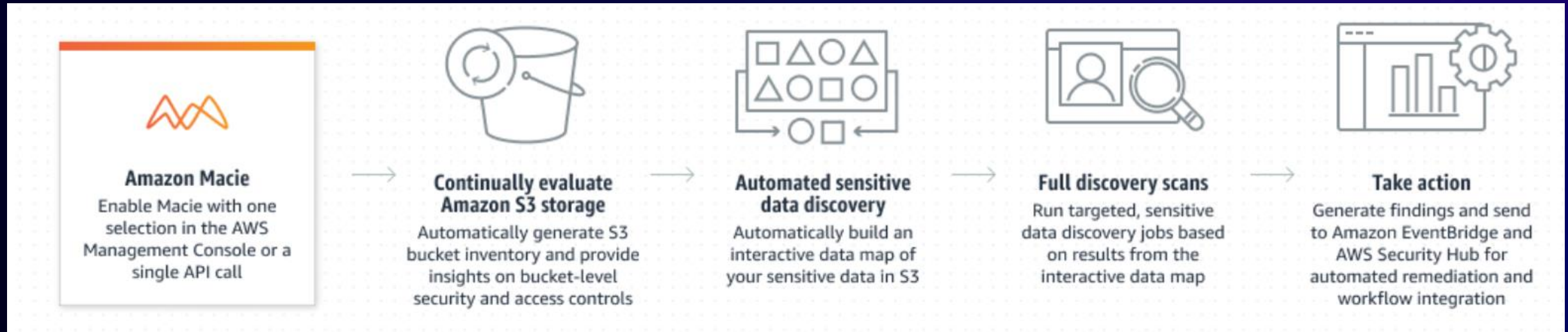
Change management

Security monitoring

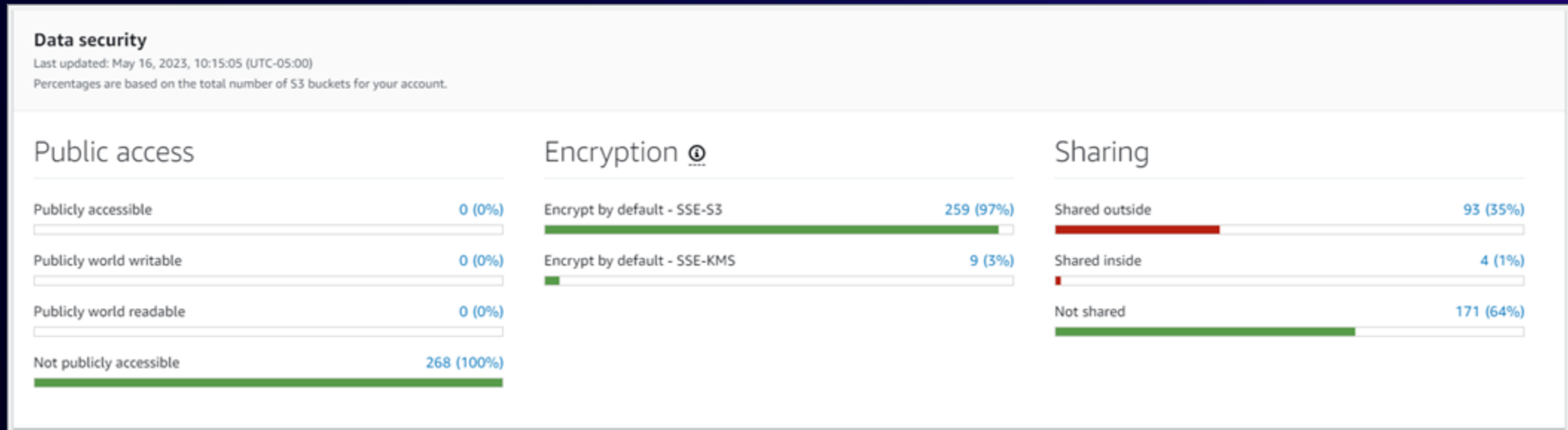
Using AWS Config to remediate KMS Keys



Amazon Macie - Overview



Amazon Macie – Example 1



Amazon Macie – Example 2

Objects by encryption type		
Customer managed	0	
SSE-KMS managed	91581	
SSE-S3 managed	326400	
No encryption	0	
Unknown	0	
Server-side encryption		
Encryption required by bucket policy	No	 
Default encryption	aws:kms	 
KMS key	arn:aws:kms:us-east-1:215724612223:key/61885e8c-10a3-4ad6-a861-494f12b7f47d	